

Table 3: Ablation study of different reward components in the RDFR training stage. Improvements are computed relative to STIT model.

Method	MSE ↓	MAE ↓
Qwen2.5-Instruct-STIT	0.403	0.428
+ R_{len}	0.401 (-0.50%)	0.425 (-0.70%)
+ R_{fmt}	0.395 (-1.99%)	0.419 (-2.10%)
+ R_{acc}	0.367 (-8.93%)	0.398 (-7.01%)
+ $R_{len} + R_{fmt}$	0.388 (-3.72%)	0.412 (-3.74%)
RDTU (R_{total})	0.351 (-12.90%)	0.382 (-10.75%)